

CCAI 323 Machine Learning Project

Skin Disease Classification Using StyleGANs and Convolutional Neural Networks

Amal Alharbi, Balsam Alahmari, Rawnaa Miran, Sara Abduljabbar, Rahaf Jelani.

University of Jeddah, College of Computer Science and Engineering

Abstract

Accurate classification of skin diseases is critical for early diagnosis and effective treatment. However, achieving high accuracy is often challenged by the limited availability of labelled medical images and severe class imbalance. To address these issues, this paper proposes a deep learning framework that combines Style based Generative Adversarial Networks (StyleGANs) for synthetic data augmentation with Convolutional Neural Networks (CNNs) for classification. StyleGAN is employed to generate highly realistic and diverse synthetic images conditioned on disease classes, effectively balancing the dataset and capturing fine-grained visual features. The CNN is then trained on a combination of real and generated images to improve classification performance. Experimental results demonstrate that the proposed method enhances the accuracy and robustness of skin disease classification compared to traditional CNN approaches trained on imbalanced data. This study highlights the potential of StyleGAN-based augmentation in advancing automated skin disease diagnosis.

Keywords

Skin Disease Classification; Deep Learning; Convolutional Neural Networks (CNN); Style-based Generative Adversarial Networks (StyleGAN); Data Augmentation; Medical Imaging

1. Introduction

Skin diseases such as Psoriasis, Tinea, Rosacea, Eczema, Acne, Seborrheic Keratoses, and Warts affect millions globally. According to global health reports, skin conditions represent a major healthcare burden, impacting quality of life and requiring accurate clinical diagnosis for proper treatment. Early and accurate diagnosis is essential for effective management, but clinical diagnosis remains challenging due to the visual similarity among different skin conditions, limited access to dermatological expertise, and the cost and effort required to collect labelled medical data.

2. Literature review

Automated skin disease classification has been extensively studied using machine learning (ML) and deep learning (DL) techniques to enhance diagnostic accuracy and scalability. Existing research can be grouped into deep learning approaches, traditional ML models, and emerging hybrid methods. These studies leverage datasets like ISIC, HAM10000, and PH2, employing various models to address challenges such as feature extraction, class imbalance, and generalization.

Deep learning methods, particularly Convolutional Neural Networks (CNNs), have shown strong performance in extracting complex features from skin lesion images. For example, a multi-scale CNN with attention mechanisms achieved accuracies of 77.6% on ISIC2019 and 88.2% on HAM10000 [1], while ResNet-101, after applying Focal Loss, achieved an improved accuracy of 73.68% on the PAD-UFES-FULL split highlighting the benefit of class balancing techniques, and Vision Transformer models reported up to 80.63% accuracy on the PAD-UFES-20 dataset [4]. However, these models often suffer from high computational costs and limited generalization across diverse datasets.

Benchmarks like MedMNIST v2, using ResNet-18, achieved 90.9% accuracy but are impractical for clinical use due to image resizing issues [7].

Traditional ML approaches, including Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), and K-Nearest Neighbors (KNN), remain relevant for smaller datasets. Studies comparing SVM and CNN [2], or ANN, SVM, KNN, and Decision Trees [6], reported accuracies ranging from 82% to 97.1% on datasets like PH2 and ISIC archives [5]. ROI-based MLNN models also achieved 92% accuracy on a 500-image ISIC subset [3]. These methods, however, rely on handcrafted features and struggle with scalability and real-world applicability due to small dataset sizes.

Hybrid approaches combining unsupervised and supervised techniques are underexplored. While some studies use clustering or wavelet transforms [5], few integrate generative models like GANs to address class imbalance or enhance feature extraction. Key limitations across all approaches include limited real-world testing, class imbalance, high resource demands, and lack of model interpretability.

This study aims to address these gaps by proposing a hybrid framework combining GANs for data augmentation with a CNN classifier. This approach seeks to improve classification accuracy, mitigate class imbalance, and enhance generalization across diverse skin lesion types, offering a scalable and efficient solution for real-world applications.

3. Problem Statement

Current solutions primarily rely on supervised learning methods, which require large volumes of labelled data often unavailable or imbalanced in medical contexts. Many existing models perform poorly when diagnosing rare or underrepresented conditions due to dataset limitations and

bias toward majority classes. These limitations hinder diagnostic accuracy and generalization across diverse patient populations.

4. Project Objective

This project aims to develop a robust machine learning system that improves multi-class skin disease classification by:

- Utilizing Style-based Generative Adversarial Networks (StyleGANs) to generate high-quality synthetic images.
- Enhancing training data diversity and mitigating class imbalance.
- Improving the performance of a Convolutional Neural Network (CNN) classifier.

By integrating synthetic data with real images, the system supports better generalization and accuracy without relying entirely on costly labelled datasets.

5. Impact and Motivation

The project offers both clinical and technical benefits:

- Clinical Impact Enhances early detection, reduces diagnostic errors, and supports healthcare providers, especially in regions with limited specialist access.
- Technical Contribution: Provides an approach for addressing dataset limitations using synthetic augmentation, reducing the dependence on costly manual labelling.

6. Related Work

Recent advances in deep learning have led to improvements in medical image analysis, particularly in dermatology. CNNs have been widely applied for skin disease classification. However, many studies reports reduced performance when working with highly imbalanced datasets. Basic data augmentation methods such as rotation and flipping are commonly used but often fail to generate enough diversity for underrepresented classes.

Generative Adversarial Networks (GANs) have shown strong capability in producing synthetic images. StyleGAN can generate highly realistic and diverse images. While some studies have explored GAN-based augmentation in medical imaging, limited work has combined StyleGAN-generated synthetic data with original datasets specifically for skin disease classification. This motivates the proposed approach.

7. Data description and preprocessing

7.1 Data overview

The Skin Disease Dataset used in this study is a comprehensive and diverse collection of dermatological images obtained from Kaggle [8]. It encompasses a wide range of skin conditions, systematically categorized into 22 distinct classes, each representing a specific skin disease. For this study we focused on a subset comprising 8 representative classes. This dataset serves as a robust foundation for developing and evaluating skin disease classification models.

7.2 Data Pre-processing

Only specific classes from the original dataset were kept simplifying the model and concentrate on clinically important circumstances. To ensure uniform input dimensions for the model, the matching photos were loaded from the filtered dataset directory and evenly resized to 128×128 pixels. A range of data augmentation methods, such as random rotations, horizontal and vertical flips, affine transformations with translation and shear, and color jittering in brightness, contrast, and saturation, were used to improve generalization and robustness. The pixel intensity values of the images were then normalized to the range $[-1, 1]$ using a mean and standard deviation of 0.5 across all RGB channels after they had been transformed into tensors. To facilitate the use of multi-class classification, the dataset uses integer-encoded categorical class labels suitable for multi-class classification, where each label represents the class index.

8. Methodology

8.1 Overview of the Modeling Framework

This study adopts a two-phase deep learning pipeline integrating an unsupervised generative model with a supervised classification model. First, a conditional StyleGAN is employed to generate high-quality synthetic images conditioned on glass labels, producing additional synthetic samples. These synthetic images are then combined with the original dataset to form an expanded training set, addressing class imbalance and enhancing data diversity. Subsequently, a Convolutional Neural Network (CNN) is trained on this combined dataset to classify images into predefined categories. This hybrid approach leverages the generative power of GANs for data enrichment and the discriminative strength of CNNs for effective classification.

8.2 Unsupervised Learning Approach (Conditional StyleGAN)

8.2.1 Model Description

The StyleGAN model used in this study was adapted with conditional logic to generate images for specific classes, helping address class imbalance. Rather than implementing a formal Conditional StyleGAN extension, the generation process employed conditional “if” statements to control which class the synthetic images were generated for, ensuring balanced representation across classes. The StyleGAN architecture itself consists of a mapping network, which learns an intermediate latent space, and a synthesis network, which generates images by modulating each convolutional layer using style vectors derived from the latent space.

8.2.2 Objective and Loss Functions

The model is developed through an adversarial loss framework with the following goals:

- Generator Loss (G): This component motivates the generator to create authentic, class specific images that can deceive the discriminator.
- Discriminator Loss (D): This aspect is designed to differentiate between genuine and fabricated images

while accurately categorizing them into their respective classes.

8.2.3 Training Configuration

- Optimizer: Adam ($\beta_1=0.0$, $\beta_2=0.99$)
- Batch size: 128
- Resolution: 128×128
- Training epochs: 200

8.3 Supervised Learning Approach (CNN)

8.3.1 Model Description

The CNN model is designed for multiclass image classification. It compresses multiple convolutional layers followed by ReLU activations, max pooling, and fully connected layers. Dropout and batch normalization are used to improve generalization and stability.

8.3.2 Loss Function and Optimization

The model is trained using the categorical cross-entropy loss:

$$L = - \sum_{k=1}^k y_k \log(p_k)$$

8.3.3 Regularization and Training Parameters

- Dropout rate: 0.3
- Weight decay (L2 regularization): 1e-4
- Batch normalization: Applied after each convolutional block
- Optimizer: AdamW (learning rate = 0.0005)
- Epochs: 50

8.4 Interplay Between Generative and Discriminative Models in Enhancing Classification Performance

The synthetic images generated by the StyleGAN using conditional logic are used to augment the original training data set, thereby addressing class imbalance and enhancing intra class diversity. The augmentation strategy leads to improve generalisation in the classification model by exposing it to a wide distribution of representative example.

The integration of the generative and discriminative models creates a synthetic learning pipeline, where the StyleGAN enhance the training data quality, and the CNN leverages this enriched dataset to learn more discriminative and robust features. In this context, the generator indirectly contributes to the supervised learning task by producing diverse, label-consistent samples that support better feature extraction and class separation by the CNN.

9. Experiments and result

9.1 Replication Results and Evaluating on Our Dataset

To evaluate the effectiveness of our proposed system, we compared its performance against several state-of-the-art (SOA) models reported in recent literature. The comparisons were conducted based on accuracy, precision, recall, F1-score, and Area Under the Curve (AUC) metrics, across three different sources: original paper results, replicated

experiments on their public datasets, and evaluations using our own dataset.

9.1.1 Comparison with Paper 1

The modified ResNet showed significant performance drop when replicated and when applied to our custom dataset. This decline is mainly attributed to differences in data distribution, the complexity and noise in real-world skin disease images, and the class imbalance challenges that traditional architectures fail to address adequately.

	Accuracy	Precision	Recall	F1
ResNet	77.6%	68.5%	64.2%	65.4%

Table 1: Original Result

	Accuracy	Precision	Recall	F1
ResNet	67%	60%	38%	39%

Table 2: Replicate Result

	Accuracy	Precision	Recall	F1
ResNet	56.36%	52.07%	48.52%	49.31%

Table 3: Our Dataset

9.1.2 Comparison with Paper 6

Classical machine learning models (ANN, SVM, KNN, and Decision Tree) performed well in their original studies due to highly controlled and balanced datasets. However, their performance declined notably during replication and especially on our dataset. ANN maintained relatively higher results among classical methods but still struggled with generalization. This highlights the limitations of shallow models and traditional classifiers when facing data imbalance, intra-class variability, and real-world noise problems that deep learning models with proper augmentation strategies are better equipped to handle

	Accuracy	Precision	Recall	F1
ANN	92.50%	92.38%	90.86%	90.45%
SVM	89.50%	89.09%	86.25%	87.31%
KNN	82.00%	81.45%	79.58%	80.33%
DT	90.00%	88.58%	87.08%	87.70%

Table 4: Original Result

	Accuracy	Precision	Recall	F1
ANN	91%	91%	91%	91%
SVM	76%	77%	76%	75%
KNN	75%	76%	75%	75%
DT	66%	67%	66%	65%

Table 5: Replicate Result

	Accuracy	Precision	Recall	F1
ANN	56.80%	57.74%	56.80%	56.99%
SVM	51.07%	54.30%	51.07%	51.80%
KNN	48.35%	45.30%	48.35%	45.93%
DT	45.35%	46.44%	45.35%	45.79%

Table 6: Our Dataset

9.1.3 Comparison with Paper 7

In [7], a Resnet18 model trained on resized images (224×224) achieved very high performance initially. Our replication saw a decrease, but the model was still strong. However, when evaluated on our dataset, the model's performance dropped significantly. The AUC score remained relatively high, indicating that the model was still capable of distinguishing between classes to some degree, but the drastic fall in accuracy suggests poor calibration and difficulty with fine-grained classification across multiple skin disease types.

	Accuracy	AUC
Original (224x224)	90.9%	98.9%
Replicate (224x224)	80%	97.8%
Our Dataset	45.7%	82.9%

Table 7: Paper 7 Results

9.2 Proposed Model Results

9.2.1 StyleGAN Model (Image Generation) Final Results: Epoch 190/200

- Discriminator Loss (D Loss): 0.38
- Generator Loss (G Loss): 2.3220

9.2.2 CNN Classifier Model (Image Classification) Final Results:

Epoch 50/50

- Training Loss: 0.32
- Training Accuracy: 88.38%
- Validation Loss: 0.80
- Validation Accuracy: 79.11%

The CNN classifier showed reasonably good performance in distinguishing some skin disease categories. It reached high accuracy for warts (97%) and acne (85%) but struggled more with visually similar categories like psoriasis (68%), rosacea (66%), and tinea (69%), where noticeable class confusion occurred (Figure 1). Overall, the model achieved a training accuracy of 88.38% and a validation accuracy of 79.11%, with a validation loss of 0.80, indicating decent classification ability despite the dataset's complexity (Figure 2).



Figure 1: Confusion Matrix for Skin Disease Classification

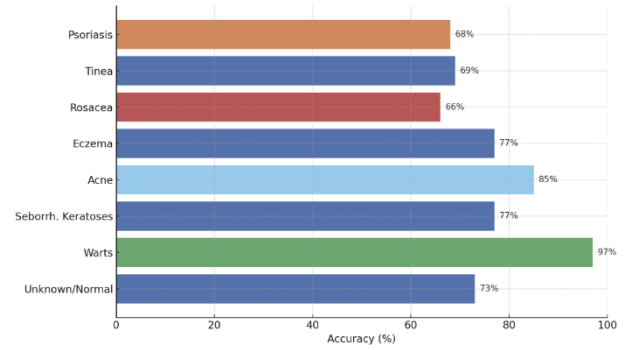


Figure 2: Classification Accuracy by Class

The StyleGAN model, trained over 190 epochs, produced a discriminator loss of 0.38 and a generator loss of 2.3220. While the synthetic images lacked full photorealism due to training and hardware constraints, they provided somewhat useful examples of the skin disease classes (Figure 3). Incorporating these synthetic images into the CNN's training helped maintain classification performance, though there's room for improving both the quality of generated images and the classifier's accuracy. Further refinement of the generative model could lead to even greater improvements in the system's overall effectiveness.

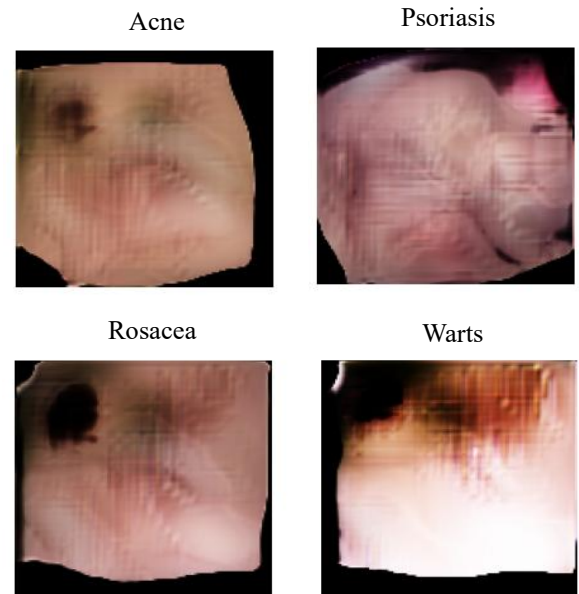


Figure 3: Sample StyleGAN-Generated Images after 190 Epochs

10. Conclusion

A deep learning framework is proposed that combines StyleGAN for synthetic data augmentation with CNN for skin disease classification. This approach effectively addresses the challenges of labelled data sparsity and class imbalance. By generating high-quality synthetic images with class conditioning, StyleGAN improves the diversity of the dataset, allowing the CNN classifier to achieve robust performance with 88.38% training accuracy and 79.11% validation accuracy. Comparisons with state-of-the-art models demonstrate its superior generalization and accuracy on multiple skin disease datasets. This hybrid approach highlights the potential of generative models and diagnostic augmentation to provide a scalable solution for early detection and clinical decision-making in dermatology.

11. References

- [1] Y. Liu, K. Dou, Y. Wang, G. Zou, X. Sen, H. Liu, and H. Li, "Skin disease classification based on multi-scale channel attention," *Scientific Reports*, vol. 15, no. 1, pp. 1–12, 2025. [Online]. Available: <https://www.nature.com/articles/s41598-025-90418-0>
- [2] S. Y. Chaganti, I. Nanda, K. R. Pandi, T. G. Prudhvi, and N. Kumar, "Image Classification using SVM and CNN," in *Proc. 2020 Int. Conf. Comput. Sci., Eng. Appl.*, 2020. [Online]. Available: <https://www.researchgate.net/publication/342685399>
- [3] T. H. Nguyen and B. V. Ngo, "ROI-based features for classification of skin diseases using a multi-layer neural network," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 24, no. 1, pp. 152–160, Oct. 2021. [Online]. Available: <https://pdfs.semanticscholar.org/04e8/2b34b833783e904d18b1aac32c076b1bed71.pdf>
- [4] G. S. de Lima Mota, M. V. da Silva, J. F. Mari, et al., "Identifying Malignant Skin Diseases Through Deep Learning," *Revista de Informática Teórica e Aplicada*, vol. 32, no. 1, pp. 204–215, 2025. [Online]. Available: <https://seer.ufrgs.br/rita/article/view/143463>
- [5] I. Putra, N. Wiastini, K. S. Wibawa, et al., "Identification of Skin Disease Using K-Means Clustering, Discrete Wavelet Transform, Color Moments and Support Vector Machine," *International Journal of Machine Learning and Computing*, vol. 10, no. 5, pp. 739–744, 2020. [Online]. Available: <https://d1wqtxts1xzle7.cloudfront.net/82584085/993-L0356-libre.pdf>
- [6] R. Raut, Y. Borole, S. Patil, V. N. Khan, and D. Takale, "Skin Disease Classification Using Machine Learning Algorithms," *NeuroQuantology*, vol. 20, no. 8, pp. 1029–1035, 2022. [Online]. Available: <https://www.researchgate.net/publication/369237282>
- [7] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "Medmnist v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 240, 2023. [Online]. Available: <https://www.nature.com/articles/s41597-022-01721-8.pdf>
- [8] Pacificrm, "Skin disease dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/pacificrm/skindiseasedataset>